

Resolución de Ecuación

Obtener constantes de A y B

$$u(t) = e^{-\zeta \omega_n t} (A \cos \omega_d t + B \sin \omega_d t)$$

$$u(0) = u_0$$

$$\dot{u}(0) = \dot{u}_0$$

PASO 1

$$A = ?$$

$$B = ?$$

$$u(0) = u_0$$

$$\dot{u}(0) = \dot{u}_0$$

| se sustituye $t=0$

$$u(0) = e^0 (A \cos 0 + B \sin 0)$$

$$e^0 = 1$$

$$\cos 0 = 1$$

$$\sin 0 = 0$$

$$\text{Se obtiene: } u(0) = 1 A(1) + 1 B(0)$$

$$u(0) = A$$

Por lo tanto

$$A = u_0$$

$$u(0) = u_0$$

PASO 2

Derivación de Soluciones.

$$f(t) = e^{-\zeta \omega_n t}$$

$$g(t) = A \cos \omega_d t + B \sin \omega_d t$$

$$\dot{u}(t) = f'(t) g(t) + f(t) g'(t)$$

$$f'(t) = -\xi \omega_n e^{-\xi \omega_n t}$$

$$g'(t) = -A \omega_d \sin \omega_d t + B \omega_d \cos \omega_d t$$

Sustituyendo

$$\dot{u}(t) = e^{-\xi \omega_n t} \left[-\xi \omega_n (A \cos(\omega_d t) + B \sin(\omega_d t)) - A \omega_d \sin(\omega_d t) + B \omega_d \cos(\omega_d t) \right]$$

Velocidad del sistema $\uparrow$

PASO 3 Aplicación de la condición inicial de la Velocidad.

$t = 0$ en la derivada

$$\dot{u}(0) = e^0 \left[-\xi \omega_n (A \cos 0 + B \sin 0) - A \omega_d \sin 0 + B \omega_d \cos 0 \right]$$

Evaluando funciones:

$$\dot{u}(0) = -\xi \omega_n A + B \omega_d$$

$$\dot{u}(0) = v_0$$

Por lo tanto

$$v_0 = -\xi \omega_n A + B \omega_d$$

ya tenemos este dato del paso 1 $\therefore A = u(0)$

Ahora $\rightarrow v_0 = -\xi \omega_n v_0 + B \omega_d$

Despejamos B $\rightarrow B \omega_d = v_0 + \xi \omega_n v_0$

$$B = \frac{V_0 + \epsilon \omega \eta V_0}{\omega d}$$

Resultado.

$$A = V_0$$

$$B = \frac{V_0 + \epsilon \omega \eta V_0}{\omega d}$$

$$U(t) = e^{-\gamma \omega t} \left[V_0 \cos(\omega d t) + \left(\frac{V_0 + \epsilon \omega \eta V_0}{\omega d} \right) \sin(\omega d t) \right]$$

